Natan Herzog

1035 Flemington St, Pittsburgh, PA 15213
201-759-6925
natan.herzog@pitt.edu

Education

Aug '20 - Present **BPhil Candidate in Engineering Science**, *Swanson School of Engineering & Frederick Hon. College*
Expected graduation in April 2025
GPA: 3.5/4.0

Certificate in Sustainability, *Mascaro Center for Sustainable Innovation*

Certificate in Engineering Simulations Design, *Swanson School of Engineering*

Certificate in Innovation, Product Design, and Entrepreneurship, *Swanson School of Engineering*

Minor in Physics, *Dietrich School of Arts and Sciences*

Research

Aug '22 - Present **OpenViewFactor**, *Applied Computational Fluid Dynamics Lab*, Dr. Matthew Barry

- Co-developing a C++ solver for calculating radiation view factors with OpenMP and CUDA for multi-core CPU and GPU acceleration
- Prototyping mesh refinement algorithms and solver interfaces in MATLAB to study convergence and failure conditions for view factor numerical integration techniques
- Developing a Python macro interface for analytic validation and OpenViewFactor solver scripting

May '22 - Aug '22 **John C. Mascaro Scholar**, *Mascaro Center for Sustainable Innovation*, Dr. David Sanchez

- Investigated the effects of various bulk nanobubble species on the oxidation potential of UV disinfection treatment of the fluorescent tracer 2'7'-dichlorofluorescin

Sept '19 - Jun '20 **Civil and Environmental Research Intern**, *New Jersey Institute of Technology*, Dr. Jay Meegoda

- Investigated the effects of Ozone nanobubbles during ultrasonic treatment of contaminated water as a new potential method for in-situ detoxification of Superfund river-sites

Working Experience

Spring '24 **Coding & ML Course Development**, *Department of Mechanical Engineering and Materials Science*

- Leading the development of a course sequence focusing on programming fundamentals and the foundations of machine learning in engineering

Summer '23 **Integration Testing Engineer**, *Ansys Inc.*

- Conducted interactive and automated testing and debugging of Ansys Additive in Mechanical with Python scripting embedded in the workbench scripting terminal

Fall '23 **Chancellor's Undergraduate Teaching Fellow**, *Statics and Mechanics of Materials 1*

- Developed a **MATLAB curriculum** to supplement in-class instruction with practical computational skills

2nd, 3rd, 4th Year **Undergraduate Teaching Assistant**

Aug '22 - Present **Makerspace Mentor - Treehouse Trainer**

Jan '22 - Present **Engineering Ambassador**

Jan '22 - Dec '22 **Engineering Science Program Student Support**

Skills

Scientific Computing	Simulation	Design/Manufacturing	Soft Skills
MATLAB	Ansys MAPDL	Solidworks/Onshape	Lateral Thinking
Python	Ansys Mechanical	3D Printing	Communication
C++	Ansys Workbench	Laser Cutting	Teamwork
CUDA	Finite Volume Method	CNC Routing	Public Speaking
CMake	Finite Difference Method	Molding/Casting	Leadership
Git	Finite Element Analysis	Prototyping	Office 365

Application of Skills

OpenViewFactor Research	Appl. Engr. Sim. in Design	Makerspace Mentoring	Comms. of Sci. & Tech.
Appl. Comp. Heat & Mass	Simulation Workshop	Infinity Mirror (W.I.P.)	Curr. Iss. in Sustainability
Adv. Engr. Applications	Ansys Co-op	Puppy Picture Frames	Society and Environment
Statics 1	Appl. Comp. Heat & Mass	Intro. Mech. Engr. Design	UTA Experience
Intro. to Thermo.	Finite Element Analysis	Statics 1	Engr. Ambassador